Lecture 3

Everything you didn't want to know about LM architecture and hyperparameters

CS336

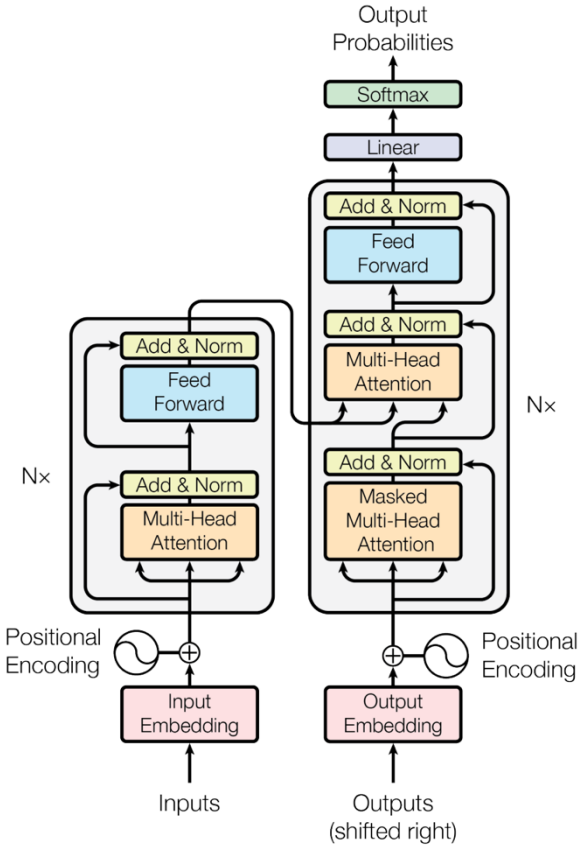
Tatsu H

Outline and goals

- Quick recap of a modern transformer (what you implement)
- What do most of the large LMs have in common?
- What are common variations to the architecture / training process?

Today's theme: the best way to learn is hands-on experience
the second best way is to try to learn from others' experience

Starting point: the ‘original’ transformer



Review: choices in the standard transformer

Position embedding: sines and cosines

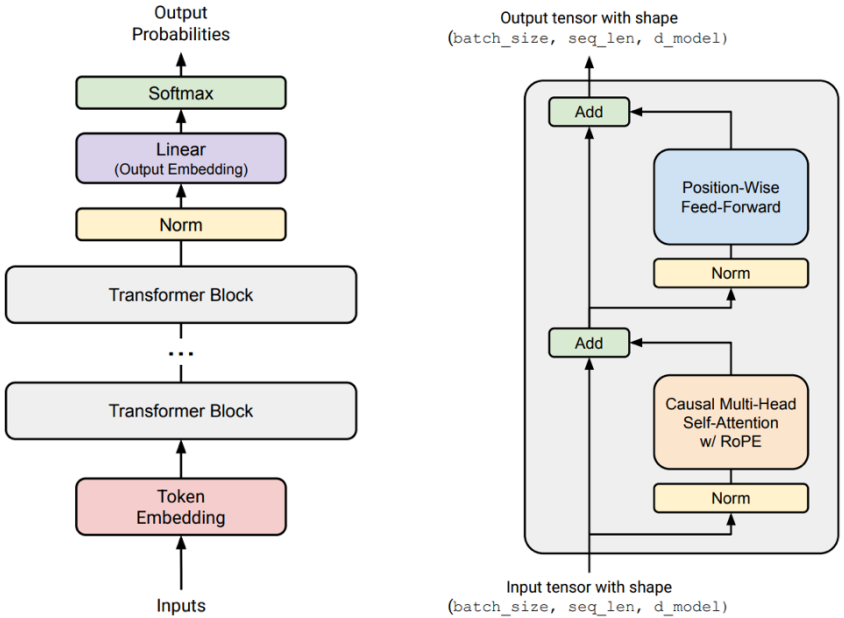
$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}})$$
$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}})$$

FFN: ReLU

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

Norm type: post-norm, LayerNorm

What you implemented – simple, modern variant



Differences:

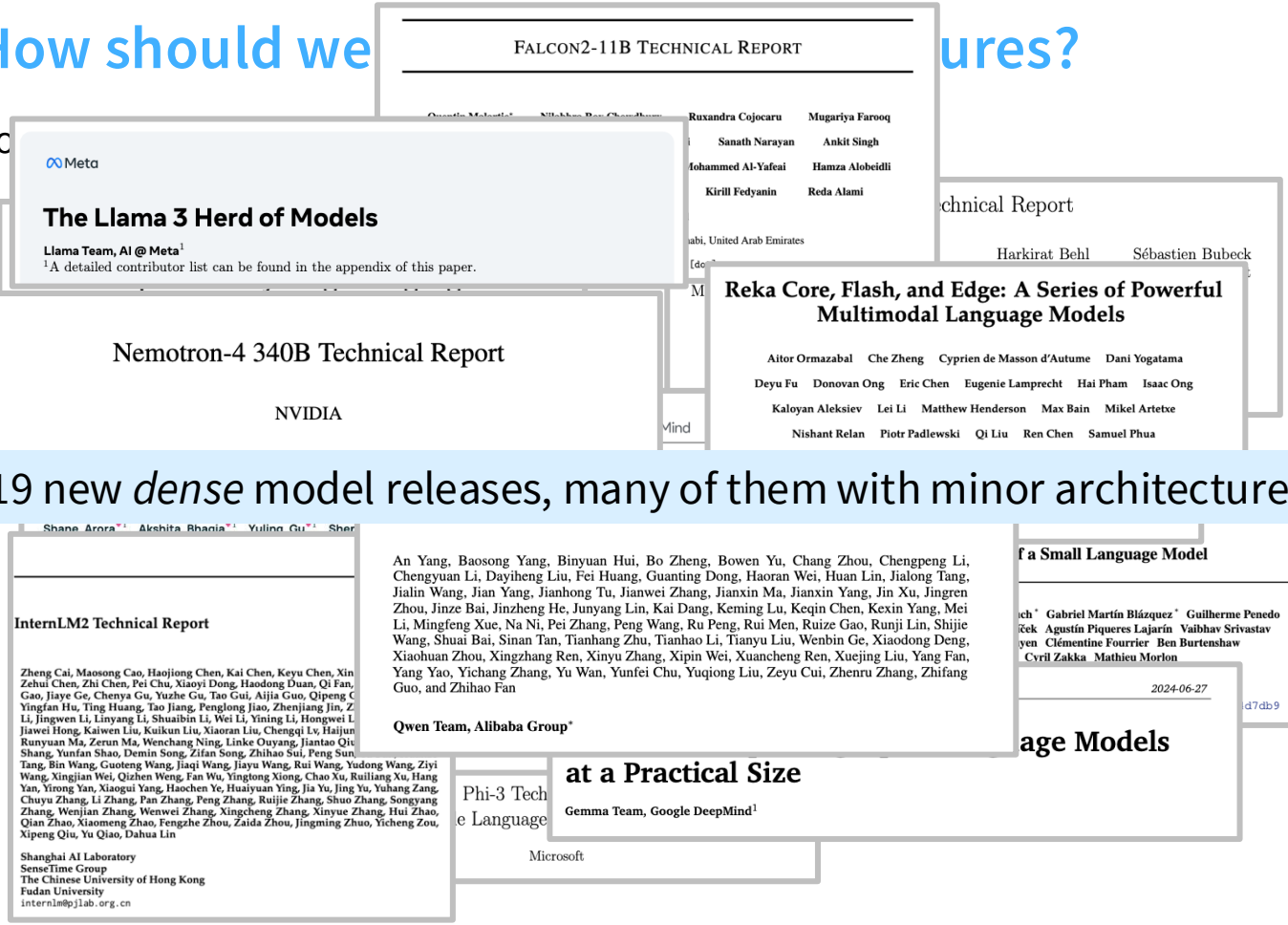
- **LayerNorm** is in front of the block
- **Rotary position embeddings (RoPE)**
- FF layers use **SwiGLU**, not ReLU
- Linear layers (and layernorm) have **no bias** (constant) terms

Why did we pick these?

What should you pick?

How should we [illegible] ures?

Lo[illegible]



FALCON2-11B TECHNICAL REPORT

Ruxandra Cojocaru Mugariya Farooq

Sanath Narayan Ankit Singh

Mohammed Al-Yafeai Hamza Alobeidli

Kirill Fedyanin Reda Alami

abi, United Arab Emirates

Meta

The Llama 3 Herd of Models

Llama Team, AI @ Meta[1]

[1]A detailed contributor list can be found in the appendix of this paper.

chnical Report

Harkirat Behl Sébastien Bubeck

Nemotron-4 340B Technical Report

NVIDIA

Reka Core, Flash, and Edge: A Series of Powerful Multimodal Language Models

Aitor Ormazabal Che Zheng Cyprien de Masson d'Autume Dani Yogatama

Deyu Fu Donovan Ong Eric Chen Eugenie Lamprecht Hai Pham Isaac Ong

Kaloyan Aleksiev Lei Li Matthew Henderson Max Bain Mikel Artetxe

Nishant Relan Piotr Padlewski Qi Liu Ren Chen Samuel Phua

Mind

Over 19 new *dense* model releases, many of them with minor architecture tweaks..

InternLM2 Technical Report

Zheng Cai, Maosong Cao, Haojiong Chen, Kai Chen, Keyu Chen, Xin
Zehui Chen, Zhi Chen, Pei Chu, Xiaoyi Dong, Haodong Duan, Qi Fan,
Gao, Jiaye Ge, Chenya Gu, Yuzhe Gu, Tao Gui, Aijia Guo, Qipeng G
Yingfan Hu, Ting Huang, Tao Jiang, Penglong Jiao, Zhenjiang Jin, Z
Li, Jingwen Li, Linyang Li, Shuaibin Li, Wei Li, Yining Li, Hongwei L
Jiawei Hong, Kaiwen Liu, Kuikun Liu, Xiaoran Liu, Chengqi Lv, Haijun
Runyuan Ma, Zerun Ma, Wenchang Ning, Linke Ouyang, Jiantao Qiu
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Tang, Bin Wang, Guoteng Wang, Jiaqi Wang, Jiayu Wang, Rui Wang, Yudong Wang, Ziyi
Wang, Xingjian Wei, Qizhen Weng, Fan Wu, Yingtong Xiong, Chao Xu, Ruiliang Xu, Hang
Yan, Yirong Yan, Xiaogui Yang, Haochen Ye, Huaiyuan Ying, Jia Yu, Jing Yu, Yuhang Zang,
Chuyu Zhang, Li Zhang, Pan Zhang, Peng Zhang, Ruijie Zhang, Shuo Zhang, Songyang
Zhang, Wenjian Zhang, Wenwei Zhang, Xingcheng Zhang, Xinyue Zhang, Hui Zhao,
Qian Zhao, Xiaomeng Zhao, Fengzhe Zhou, Zaida Zhou, Jingming Zhuo, Yicheng Zou,
Xipeng Qiu, Yu Qiao, Dahua Lin

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Qwen Team, Alibaba Group*

f a Small Language Model

Gabriel Martín Blázquez* Guilherme Penedo

Agustín Piqueres Lajarín Vaibhav Srivastav

Clémentine Fourrier Ben Burtenshaw

Cyril Zakka Mathieu Morlon

2024-06-27

d7db9

age Models

at a Practical Size

Gemma Team, Google DeepMind[1]

Phi-3 Tech

e Language

Microsoft

August 5, 2025 Release Product

Introducing gpt-oss

gpt-oss-120b and gpt-oss-20b push the frontier of open-weight reasoning model

Explore on Hugging Face ↗ Read model card ↗

MiniMax-M1: Scaling Test-Time Efficiently with Lightning Attent

MiniMax[1]

Llama 4:
Leading Multimodal In

Llama 4 Behemoth

Llama 4 Maverick

Llama 4 Scout

deepseek

DeepSeek-V3.2: Pushing the Frontier of Open Large Language Models

DeepSeek-AI

research@deepseek.com

2025-05-15

efficien
hybrid models for
enterprise

Published 02 October 2025

modal,

Liquid

Try LFM • Docs • LEAP • Disco

MiniMax M2 & Agent: Ingenious in Simplicity

Access API → Coding Plan → Try Agent Now

NVIDIA

NVIDIA Nemotron 3: Efficient and

models—Nano, Super, and Ultra. These models deliver strong
apabilities. The Nemotron 3 family uses a Mixture-of-Experts
re to provide best-in-class throughput and context lengths of
dels are trained with NVFP4 and incorporate LatentMoE, a
lity. The two larger models also include MTP layers for faster

INTERN-S1: A SCIENTIFIC MULTIMODAL FOUNDATION MODEL

Intern-S1 Team, Shanghai AI Laboratory

foundation models—together. We're
ratch, and sharing and programmatically
e, the data, the experiments, the
ite anyone who shares our vision of open
nd contribute, whether you want to try out
ithm, dataset, evaluation…there is a lot to

ERNIE 4.5 Technical Repo

KIMI K2: OPEN AGENTIC INTELLIGENCE

TECHNICAL REPORT OF KIMI K2

Kimi Team

oundation models
lixture-of-Experts
taining 424B total
MoE architecture,

of Maryland
here.

2025-12-22 · Research

GLM-4.7: Advancing the Coding Capability

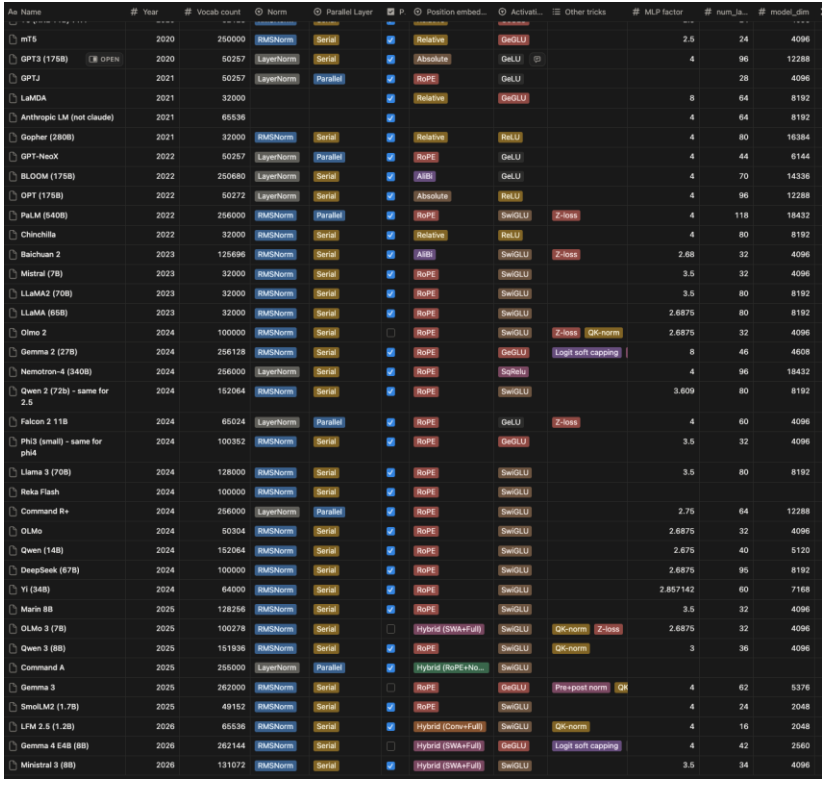
Try it at Z.ai ↗ Call it at Z.ai ↗ Z.ai Coding Plan ↗ GitHub ↗ HuggingFace ↗ Tech Report ↗

Step-3 is Large yet Affordable: Model-system Co-design for Cost-effective Decoding

StepFun Inc.

Let’s look at the data (on dense architectures)

Learn from the many other models (and papers) out there



Name	Year	Vocab count	Norm	Parallel Layer	P	Position embed...	Activati...	Other tricks	MLP factor	num_la...	model_dim
mT5	2020	250000	RMSNorm	Serial	☑	Relative	GeGLU		2.5	24	4096
GPT3 (175B) OPEN	2020	50257	LayerNorm	Serial	☑	Absolute	GeLU		4	96	12288
GPTJ	2021	50257	LayerNorm	Parallel	☑	RoPE	GeLU			28	4096
LaMDA	2021	32000			☑	Relative	GeGLU		8	64	8192
Anthropic LM (not claude)	2021	65536			☑				4	64	8192
Gopher (280B)	2021	32000	RMSNorm	Serial	☑	Relative	ReLU		4	80	16384
GPT-NeoX	2022	50257	LayerNorm	Parallel	☑	RoPE	GeLU		4	44	6144
BLOOM (175B)	2022	250680	LayerNorm	Serial	☑	AliBi	GeLU		4	70	14336
OPT (175B)	2022	50272	LayerNorm	Serial	☑	Absolute	ReLU		4	96	12288
PaLM (540B)	2022	256000	RMSNorm	Parallel	☑	RoPE	SwiGLU	Z-loss	4	118	18432
Chinchilla	2022	32000	RMSNorm	Serial	☑	Relative	ReLU		4	80	8192
Baichuan 2	2023	125696	RMSNorm	Serial	☑	AliBi	SwiGLU	Z-loss	2.68	32	4096
Mistral (7B)	2023	32000	RMSNorm	Serial	☑	RoPE	SwiGLU		3.5	32	4096
LLaMA2 (70B)	2023	32000	RMSNorm	Serial	☑	RoPE	SwiGLU		3.5	80	8192
LLaMA (65B)	2023	32000	RMSNorm	Serial	☑	RoPE	SwiGLU		2.6875	80	8192
Olmo 2	2024	100000	RMSNorm	Serial	☐	RoPE	SwiGLU	Z-loss QK-norm	2.6875	32	4096
Gemma 2 (27B)	2024	256128	RMSNorm	Serial	☑	RoPE	GeGLU	Logit soft capping	8	46	4608
Nemotron-4 (340B)	2024	256000	LayerNorm	Serial	☑	RoPE	SqReLu		4	96	18432
Qwen 2 (72b) - same for 2.5	2024	152064	RMSNorm	Serial	☑	RoPE	SwiGLU		3.609	80	8192
Falcon 2 11B	2024	65024	LayerNorm	Parallel	☑	RoPE	GeLU	Z-loss	4	60	4096
Phi3 (small) - same for phi4	2024	100352	RMSNorm	Serial	☑	RoPE	GeGLU		3.5	32	4096
Llama 3 (70B)	2024	128000	RMSNorm	Serial	☑	RoPE	SwiGLU		3.5	80	8192
Reka Flash	2024	100000	RMSNorm	Serial	☑	RoPE	SwiGLU				
Command R+	2024	256000	LayerNorm	Parallel	☑	RoPE	SwiGLU		2.75	64	12288
OLMo	2024	50304	RMSNorm	Serial	☑	RoPE	SwiGLU		2.6875	32	4096
Qwen (14B)	2024	152064	RMSNorm	Serial	☑	RoPE	SwiGLU		2.675	40	5120
DeepSeek (67B)	2024	100000	RMSNorm	Serial	☑	RoPE	SwiGLU		2.6875	95	8192
Yi (34B)	2024	64000	RMSNorm	Serial	☑	RoPE	SwiGLU		2.857142	60	7168
Marin 8B	2025	128256	RMSNorm	Serial	☑	RoPE	SwiGLU		3.5	32	4096
OLMo 3 (7B)	2025	100278	RMSNorm	Serial	☐	Hybrid (SWA+Full)	SwiGLU	QK-norm Z-loss	2.6875	32	4096
Qwen 3 (8B)	2025	151936	RMSNorm	Serial	☑	RoPE	SwiGLU	QK-norm	3	36	4096
Command A	2025	255000	LayerNorm	Parallel	☑	Hybrid (RoPE+No...	SwiGLU				
Gemma 3	2025	262000	RMSNorm	Serial	☐	RoPE	GeGLU	Pre+post norm QK	4	62	5376
SmolLM2 (1.7B)	2025	49152	RMSNorm	Serial	☑	RoPE	SwiGLU		4	24	2048
LFM 2.5 (1.2B)	2026	65536	RMSNorm	Serial	☑	Hybrid (Conv+Full)	SwiGLU	QK-norm	4	16	2048
Gemma 4 E4B (8B)	2026	262144	RMSNorm	Serial	☐	Hybrid (SWA+Full)	GeGLU	Logit soft capping	4	42	2560
Ministral 3 (8B)	2026	131072	RMSNorm	Serial	☑	Hybrid (SWA+Full)	SwiGLU		3.5	34	4096

We will talk through many major architecture and hyperparameter variants.

- What do all these models have in common?
- What parts vary?
- What can we learn from this?

What are we going to cover?

Common architecture variations

- Activations, FFN
- Attention variants
- Position embeddings

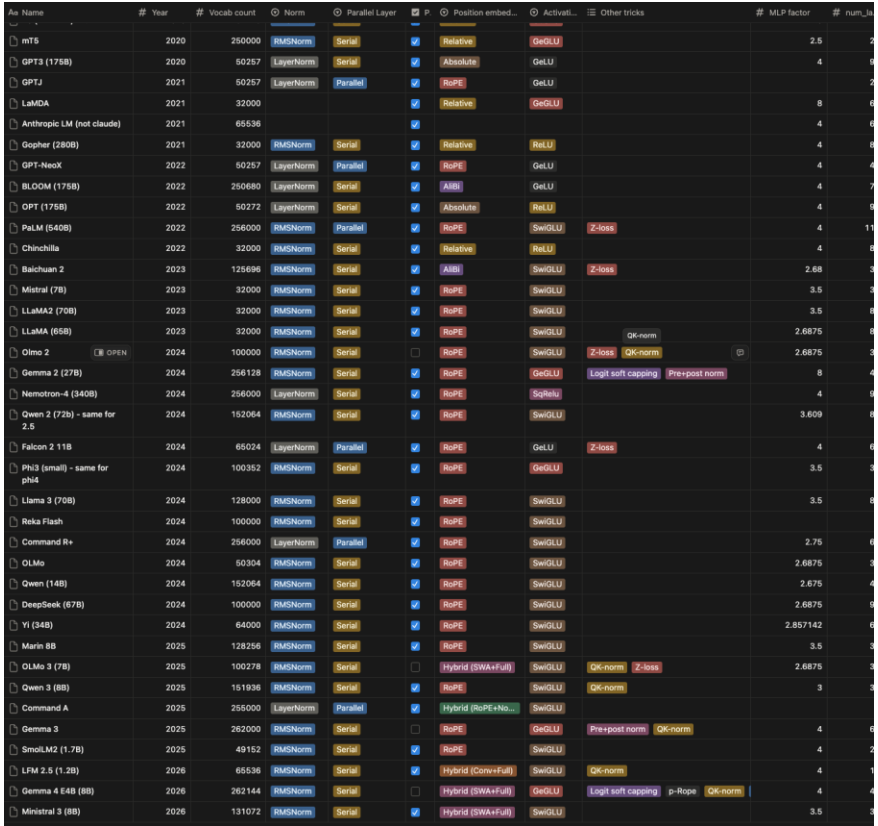
Hyperparameters that (do or don’t) matter

- What is ff_dim? Do multi_head dims always sum to model_dim?
- How many vocab elements?

Stability tricks

Architecture variations..

Let’s think about the core architecture piece



Name	Year	Vocab count	Norm	Parallel Layer	P.	Position embed...	Activati...	Other tricks	MLP factor	num_la
mT5	2020	250000	RMSNorm	Serial	✓	Relative	GeGLU		2.5	
GPT3 (175B)	2020	50257	LayerNorm	Serial	✓	Absolute	GeLU		4	
GPTJ	2021	50257	LayerNorm	Parallel	✓	RoPE	GeLU			
LaMDA	2021	32000			✓	Relative	GeGLU		8	
Anthropic LM (not claude)	2021	65536			✓				4	
Gopher (280B)	2021	32000	RMSNorm	Serial	✓	Relative	ReLU		4	
GPT-NeoX	2022	50257	LayerNorm	Parallel	✓	RoPE	GeLU		4	
BLOOM (175B)	2022	250680	LayerNorm	Serial	✓	ALiBi	GeLU		4	
OPT (175B)	2022	50272	LayerNorm	Serial	✓	Absolute	ReLU		4	
PaLM (540B)	2022	256000	RMSNorm	Parallel	✓	RoPE	SwiGLU	Z-loss	4	
Chinchilla	2022	32000	RMSNorm	Serial	✓	Relative	ReLU		4	
Baichuan 2	2023	125696	RMSNorm	Serial	✓	ALiBi	SwiGLU	Z-loss	2.68	
Mistral (7B)	2023	32000	RMSNorm	Serial	✓	RoPE	SwiGLU		3.5	
LLaMA2 (70B)	2023	32000	RMSNorm	Serial	✓	RoPE	SwiGLU		3.5	
LLaMA (65B)	2023	32000	RMSNorm	Serial	✓	RoPE	SwiGLU	QK-norm	2.6875	
Olmo 2 OPEN	2024	100000	RMSNorm	Serial		RoPE	SwiGLU	Z-loss QK-norm	2.6875	
Gemma 2 (27B)	2024	256128	RMSNorm	Serial	✓	RoPE	GeGLU	Logit soft capping Pre+post norm	8	
Nemotron-4 (340B)	2024	256000	LayerNorm	Serial	✓	RoPE	SqRelu		4	
Qwen 2 (72b) - same for 2.5	2024	152064	RMSNorm	Serial	✓	RoPE	SwiGLU		3.609	
Falcon 2 11B	2024	65024	LayerNorm	Parallel	✓	RoPE	GeLU	Z-loss	4	
Phi3 (small) - same for phi4	2024	100352	RMSNorm	Serial	✓	RoPE	GeGLU		3.5	
Llama 3 (70B)	2024	128000	RMSNorm	Serial	✓	RoPE	SwiGLU		3.5	
Reka Flash	2024	100000	RMSNorm	Serial	✓	RoPE	SwiGLU			
Command R+	2024	256000	LayerNorm	Parallel	✓	RoPE	SwiGLU		2.75	
OLMo	2024	50304	RMSNorm	Serial	✓	RoPE	SwiGLU		2.6875	
Qwen (14B)	2024	152064	RMSNorm	Serial	✓	RoPE	SwiGLU		2.675	
DeepSeek (67B)	2024	100000	RMSNorm	Serial	✓	RoPE	SwiGLU		2.6875	
Yi (34B)	2024	64000	RMSNorm	Serial	✓	RoPE	SwiGLU		2.857142	
Marin 8B	2025	128256	RMSNorm	Serial	✓	RoPE	SwiGLU		3.5	
OLMo 3 (7B)	2025	100278	RMSNorm	Serial		Hybrid (SWA+Full)	SwiGLU	QK-norm Z-loss	2.6875	
Qwen 3 (8B)	2025	151936	RMSNorm	Serial	✓	RoPE	SwiGLU	QK-norm	3	
Command A	2025	255000	LayerNorm	Parallel	✓	Hybrid (RoPE+No...	SwiGLU			
Gemma 3	2025	262000	RMSNorm	Serial		RoPE	GeGLU	Pre+post norm QK-norm	4	
SmolLM2 (1.7B)	2025	49152	RMSNorm	Serial	✓	RoPE	SwiGLU		4	
LFM 2.5 (1.2B)	2026	65536	RMSNorm	Serial	✓	Hybrid (Conv+Full)	SwiGLU	QK-norm	4	
Gemma 4 E4B (8B)	2026	262144	RMSNorm	Serial		Hybrid (SWA+Full)	GeGLU	Logit soft capping p-Rope QK-norm	4	
Ministral 3 (8B)	2026	131072	RMSNorm	Serial	✓	Hybrid (SWA+Full)	SwiGLU		3.5	

High level view:

- Dominance of ‘LLaMA-like’ architectures
- Trends over the years (QK-norm, Hybrid attention)

Pre-vs-post norm

The one thing *everyone* agrees on (in 2024)

大家一致同意：prenorm更好一些
不过这具体是为啥呢

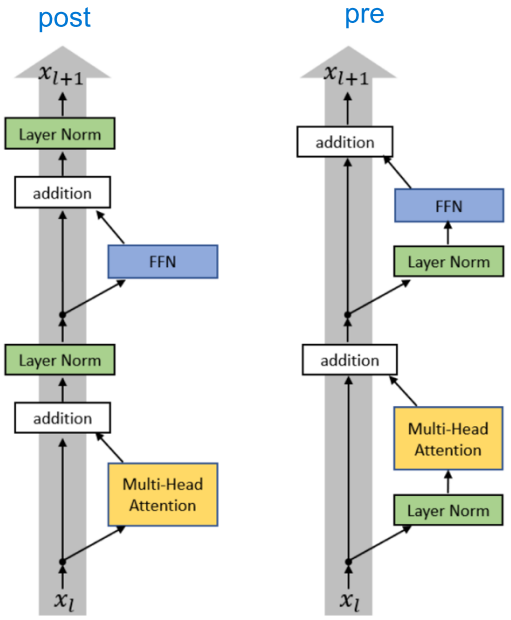


Figure from Xiong 2020

Post-LN Transformer	Pre-LN Transformer
$x_{l,i}^{post,1} = \text{MultiHeadAtt}(x_{l,i}^{post}, [x_{l,1}^{post}, \cdots, x_{l,n}^{post}])$	$x_{l,i}^{pre,1} = \text{LayerNorm}(x_{l,i}^{pre})$
$x_{l,i}^{post,2} = x_{l,i}^{post} + x_{l,i}^{post,1}$	$x_{l,i}^{pre,2} = \text{MultiHeadAtt}(x_{l,i}^{pre,1}, [x_{l,1}^{pre,1}, \cdots, x_{l,n}^{pre,1}])$
$x_{l,i}^{post,3} = \text{LayerNorm}(x_{l,i}^{post,2})$	$x_{l,i}^{pre,3} = x_{l,i}^{pre} + x_{l,i}^{pre,2}$
$x_{l,i}^{post,4} = \text{ReLU}(x_{l,i}^{post,3} W^{1,l} + b^{1,l}) W^{2,l} + b^{2,l}$	$x_{l,i}^{pre,4} = \text{LayerNorm}(x_{l,i}^{pre,3})$
$x_{l,i}^{post,5} = x_{l,i}^{post,3} + x_{l,i}^{post,4}$	$x_{l,i}^{pre,5} = \text{ReLU}(x_{l,i}^{pre,4} W^{1,l} + b^{1,l}) W^{2,l} + b^{2,l}$
$x_{l+1,i}^{post} = \text{LayerNorm}(x_{l,i}^{post,5})$	$x_{l+1,i}^{pre} = x_{l,i}^{pre,5} + x_{l,i}^{pre,3}$
	Final LayerNorm: $x_{Final,i}^{pre} \leftarrow \text{LayerNorm}(x_{L+1,i}^{pre})$

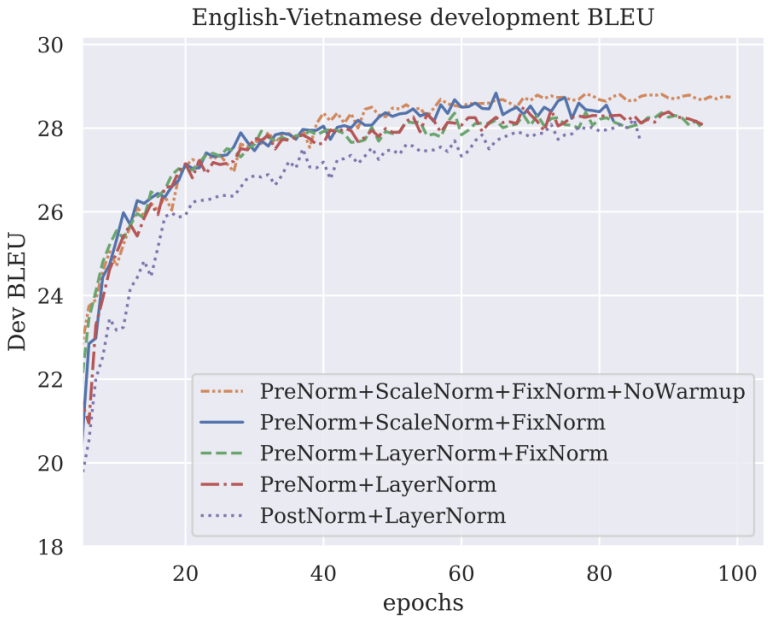
Set up LayerNorm so that it doesn't affect the main residual signal path (on the left)

灰色代表residualstream
postnorm也叫做residual norm

Almost all modern LMs use pre-norm (but BERT was post-norm)

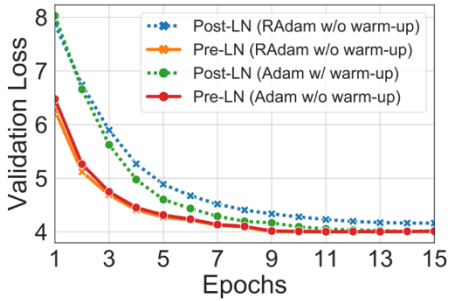
(One somewhat funny exception – OPT350M. I don't know why this is post-norm)

Pre-vs-post-norm, the data

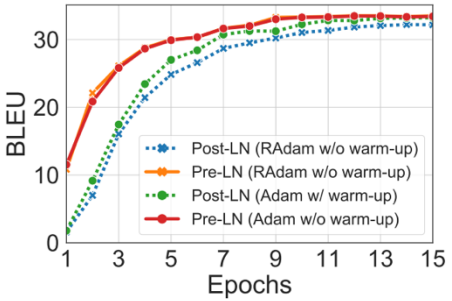


最初的想法：prenorm让LLM的训练更好地收敛

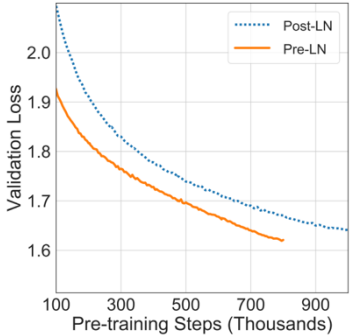
Salazar and Ngyuen 2019



(a) Validation Loss (IWSLT)



(b) BLEU (IWSLT)



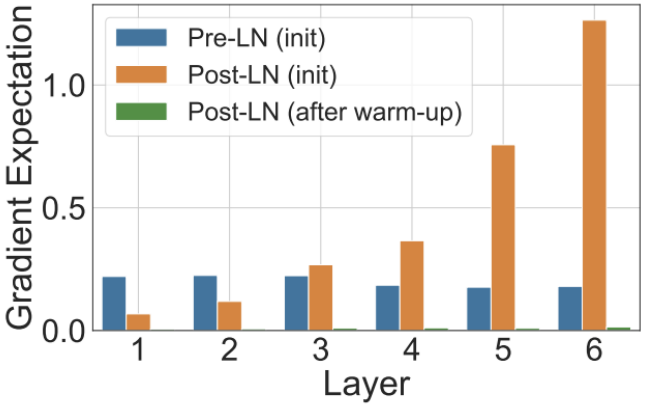
(a) Validation Loss on BERT

Figure from Xiong 2020

Pre-vs-post norm, explanations?

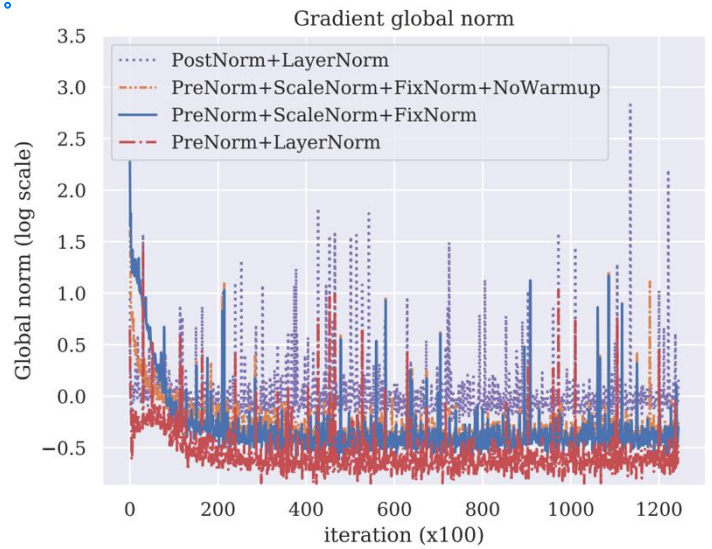
Gradient attenuation [Xiong 2020]

#TODO：这里不是很理解为啥post-LN会使得梯度不稳定，而pre-LN就不会。



(a) W^1 in the FFN sub-layers

Gradient spikes [Salazar and Ngyuen]



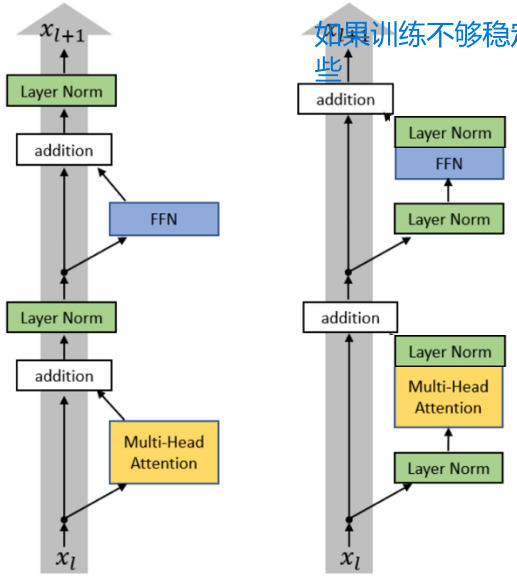
一个想法：keep the residual stream clean，这样反向传播的时候，才能真正发挥residual的作用

Original stated advantage– removing warmup.

Today – stability and larger LRs for large networks

New things – ‘double’ norm or non-residual postnorm

If putting LayerNorms in residual streams is bad.. Why not post-norm outside the stream?



如果训练不够稳定，那么就多加几个layernorm，然后训练就会稳定一些

Recent models: Grok, Gemma 2. Olmo 2 *only* does non-residual post norm

LayerNorm vs RMSNorm

Original transformer: **LayerNorm** – normalizes the mean and variance across d_{model}

$$y = \frac{x - \mathrm{E}[x]}{\sqrt{\mathrm{Var}[x] + \epsilon}} * \gamma + \beta$$

Notable models:

GPT3/2/1, OPT, GPT-J, BLOOM

Many modern LMs: **RMSNorm** – does not subtract mean or add a bias term

$$y = \frac{x}{\sqrt{||x||_2^2 + \varepsilon}} * \gamma$$

Notable models:

LLaMA-family, PaLM, Chinchilla, T5

#TODO 详细的layernorm公式，配合具体batch的样本形状变换和计算说明，以及和RMSNorm真正本质的区别

Why RMSNorm?

Modern explanation – it's faster (and just as good).

- **Fewer operations** (no mean calculation)
- **Fewer parameters** (no bias term to store)

$$y = \frac{x - \mathrm{E}[x]}{\sqrt{\mathrm{Var}[x] + \epsilon}} * \gamma + \beta$$

Does this explanation make sense?

Operator class	% flop
△ Tensor contraction	99.80
□ Stat. normalization	0.17
○ Element-wise	0.03

mean和bias这些运算其实对于我们的效用不高，那么就移除这些运算

Matrix multiplies are the *vast* majority of FLOPs (and memory)

[Ivanov et al 2023]

Why RMSNorm (2)

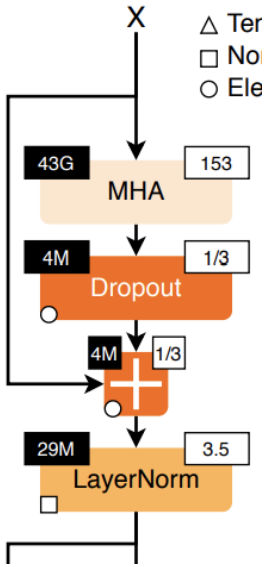
Important lesson: FLOPS are not runtime! (we will discuss this in far more detail later)

layernorm的运行flops不大，但是运行耗时（runtime）很长

Operator class	% flop	% Runtime
△ Tensor contraction	99.80	61.0
□ Stat. normalization	0.17	25.5
○ Element-wise	0.03	13.5

RMSNorm can still matter due to the importance of *data movement*

data movementL，也就是将data从HMB移动到加速单元，是很耗时的



Left top (”43G”) is FLOPS
Right top (“153”) is the FLOP-to-memory ratio

[Ivanov et al 2023]

RMSNorm - validation

RMSNorm runtime (and surprisingly, perf) gains have been seen in papers

Model	Params	Ops	Step/s	Early loss	Final loss	SGLUE	XSum	WebQ	WMT EnDe
Vanilla Transformer	$223M$	$11.1T$	3.50	2.182 ± 0.005	1.838	71.66	17.78	23.02	26.62
RMS Norm	$223M$	$11.1T$	3.68	2.167 ± 0.008	**1.821**	**75.45**	**17.94**	**24.07**	**27.14**
Rezero	$223M$	$11.1T$	3.51	2.262 ± 0.003	1.939	61.69	15.64	20.90	26.37
Rezero + LayerNorm	$223M$	$11.1T$	3.26	2.223 ± 0.006	1.858	70.42	17.58	23.02	26.29
Rezero + RMS Norm	$223M$	$11.1T$	3.34	2.221 ± 0.009	1.875	70.33	17.32	23.02	26.19
Fixup	$223M$	$11.1T$	2.95	2.382 ± 0.012	2.067	58.56	14.42	23.02	26.31

Narang et al 2020

More generally: dropping bias terms

Most modern transformers don't have bias terms.

bias在transformers架构中其实没那么重要

可以剔除

Original Transformer:

$$\mathrm{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

Most implementations (if they're not gated):

$$FFN(x) = \sigma(xW_1)W_2$$

Reasons: memory (similar to RMSnorm) and optimization stability

LayerNorm: recap

- Basically everyone does non-residual norm (often prenorm)
  - Intuition – keep the good parts of residual connections
  - Observations – nicer gradient propagation, fewer spike
  - Some people add a second norm outside the residual stream

- Most people do RMSnorm
  - In practice, works as well as LayerNorm
  - But, has fewer parameters to move around, which saves on wallclock time
  - People more generally drop bias terms since the compute/param tradeoffs are not great.

Activations

A whole zoo of activations ..

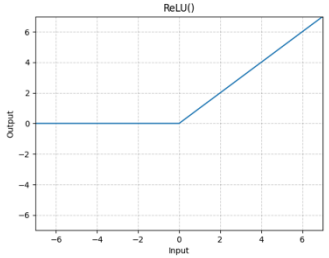
ReLU, GeLU, Swish, ELU, GLU, GeGLU, ReGLU, SeLU, SwiGLU, LiGLU

What are these things? What do people use? Does it matter?

A few of the common activations

ReLU

$$FF(x) = \max(0, xW_1)\, W_2$$

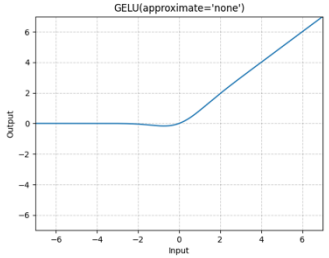


Notable models:

Original transformer, T5, Gopher, Chinchilla, OPT

GeLU

$$FF(x) = \mathrm{GELU}(xW_1)W_2$$
$$GELU(x) \coloneqq x\Phi(x)$$



Notable models:

GPT1/2/3, GPTJ, GPT-Neox, BLOOM

SwiGLU / GeGLU (next slide..)

Notable models:

Llama, PaLM,T5 v1.1, *most* models post 2023

Gated activations (*GLU)

门控激活函数

GLUs modify the ‘first part’ of a FF layer

$$FF(x) = \max(0, xW_1)\, W_2$$

Instead of a linear + ReLU, augment the above with an (entrywise) linear term

$$\max(0, xW_1) \rightarrow \max(0, xW_1) \otimes (xV)$$

This gives the gated variant (ReGLU) – note that we have an extra parameter (V)

$$\mathrm{FF}_{\mathrm{ReGLU}}(x) = (\max(0, xW_1) \otimes xV)\, W_2$$

Gated variants of standard FF layers

GeGLU

$$\mathrm{FFN}_{\mathrm{GEGLU}}(x, W, V, W_2) = (\mathrm{GELU}(xW) \otimes xV)W_2$$

Notable models:

T5 v1.1, mT5, LaMDA, Phi3, Gemma 2, Gemma 3, Gemma 4

SwiGLU (swish is $x * \mathrm{sigmoid}(x)$)

$$\mathrm{FFN}_{\mathrm{SwiGLU}}(x, W, V, W_2) = (\mathrm{Swish}_1(xW) \otimes xV)W_2$$

Notable models:

LLaMa 1/2/3, PaLM, Mistral, OlMo, *most models post 2023*

Note: Gated models use smaller dimensions for the d_{ff} by 2/3 控制总参数量一致

Do gated linear units work?

Yes, fairly consistently so.

	Score Average	CoLA MCC	SST-2 Acc
FFN_{ReLU}	83.80	51.32	94.04
FFN_{GELU}	83.86	53.48	94.04
$\text{FFN}_{\text{Swish}}$	83.60	49.79	93.69
FFN_{GLU}	84.20	49.16	94.27
$\text{FFN}_{\text{GEGLU}}$	84.12	53.65	93.92
$\text{FFN}_{\text{Bilinear}}$	83.79	51.02	**94.38**
$\text{FFN}_{\text{SwiGLU}}$	84.36	51.59	93.92
$\text{FFN}_{\text{ReGLU}}$	**84.67**	**56.16**	**94.38**
[Raffel et al., 2019]	83.28	53.84	92.68
ibid. stddev.	0.235	1.111	0.569

Shazeer 2020

Do gated linear units work (2)?

Yes, with other works corroborating Shazeer 2020

Model	Params	Ops	Step/s	Early loss	Final loss	SGLUE	XSum	WebQ
Vanilla Transformer	$223M$	$11.1T$	3.50	2.182 ± 0.005	1.838	71.66	17.78	23.02
GeLU	$223M$	$11.1T$	3.58	2.179 ± 0.003	1.838	**75.79**	**17.86**	**25.13**
Swish	$223M$	$11.1T$	3.62	2.186 ± 0.003	1.847	**73.77**	17.74	**24.34**
ELU	$223M$	$11.1T$	3.56	2.270 ± 0.007	1.932	67.83	16.73	23.02
GLU	$223M$	$11.1T$	3.59	2.174 ± 0.003	**1.814**	**74.20**	**17.42**	24.34
GeGLU	$223M$	$11.1T$	3.55	2.130 ± 0.006	**1.792**	**75.96**	**18.27**	**24.87**
ReGLU	$223M$	$11.1T$	3.57	2.145 ± 0.004	**1.803**	**76.17**	**18.36**	**24.87**
SeLU	$223M$	$11.1T$	3.55	2.315 ± 0.004	1.948	68.76	16.76	22.75
SwiGLU	$223M$	$11.1T$	3.53	2.127 ± 0.003	**1.789**	**76.00**	**18.20**	**24.34**
LiGLU	$223M$	$11.1T$	3.59	2.149 ± 0.005	**1.798**	**75.34**	**17.97**	**24.34**
Sigmoid	$223M$	$11.1T$	3.63	2.291 ± 0.019	1.867	**74.31**	17.51	23.02
Softplus	$223M$	$11.1T$	3.47	2.207 ± 0.011	1.850	**72.45**	17.65	**24.34**

Narang et al 2020

Gating, activations

- **Many variations (ReLU, GeLU, *GLU) across models.**

- ***GLU isn't *necessary* for a working model (see GPT3), but it's rare to see others..**

  Some outlier models..

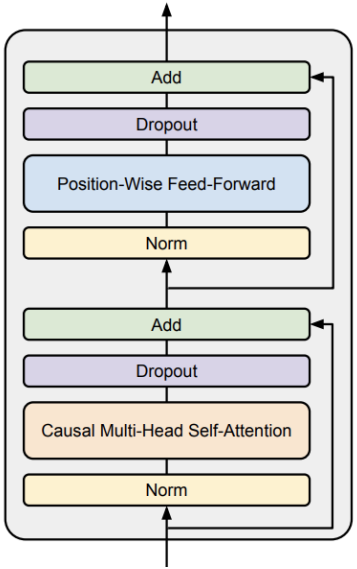
  Nemotron 340B (Squared ReLU)

基本所有模型都用了Gated的激活函数

- **Evidence points towards somewhat consistent gains from Swi/GeGLU**

Serial vs Parallel layers

Normal transformer blocks are *serial* – they compute attention, then the MLP



Could we parallelize the transformer block?

Parallel layers

TODO parallel是啥？和经典的serial架构区别？就使用这个GPT-J的架构具体说明

A few models (GPTJ, PaLM, GPT-NeoX) do parallel layers. Originally in GPT-J

> **Parallel Layers** – We use a "parallel" formulation in each Transformer block (Wang & Komatsuzaki, 2021), rather than the standard "serialized" formulation. Specifically, the standard formulation can be written as:
>
> $$y = x + \text{MLP}(\text{LayerNorm}(x + \text{Attention}(\text{LayerNorm}(x))))$$
>
> Whereas the parallel formulation can be written as:
>
> $$y = x + \text{MLP}(\text{LayerNorm}(x)) + \text{Attention}(\text{LayerNorm}(x))$$
>
> The parallel formulation results in roughly 15% faster training speed at large scales, since the MLP and Attention input matrix multiplications can be fused. Ablation experiments showed a small quality degradation at 8B scale but no quality degradation at 62B scale, so we extrapolated that the effect of parallel layers should be quality neutral at the 540B scale.

If implemented right, LayerNorm can be shared, and matrix multiplies can be fused

Recent Models: Cohere Command A, Falcon 2 11B, Command R+

Summary: architectures

Pre-vs-post norm:

- Everyone does non-residual norm (except OPT350M), likely with good reason.

Layer vs RMSnorm:

- RMSnorm has clear compute wins, sometimes even performance

Gating:

- GLUs are consensus now 共识

Serial vs parallel layers:

- Most models now use serial layers

Name	Year	Norm	Parallel Layer	Pre-norm	Position embedding	Activations
Original transformer	2017	LayerNorm	Serial	☐	Sine	ReLU
GPT	2018	LayerNorm	Serial	☐	Absolute	GeLU
T5 (11B)	2019	RMSNorm	Serial	☑	Relative	ReLU
GPT2	2019	LayerNorm	Serial	☑	Absolute	GeLU
T5 (XXL 11B) v1.1	2020	RMSNorm	Serial	☑	Relative	GeGLU
mT5	2020	RMSNorm	Serial	☑	Relative	GeGLU
GPT3 (175B)	2020	LayerNorm	Serial	☑	Absolute	GeLU
GPTJ	2021	LayerNorm	Parallel	☑	RoPE	GeLU
LaMDA	2021			☑	Relative	GeGLU
Anthropic LM (not claude)	2021			☑		
Gopher (280B)	2021	RMSNorm	Serial	☑	Relative	ReLU
GPT-NeoX	2022	LayerNorm	Parallel	☑	RoPE	GeLU
BLOOM (175B)	2022	LayerNorm	Serial	☑	AliBi	GeLU
OPT (175B)	2022	LayerNorm	Serial	☑	Absolute	ReLU
PaLM (540B)	2022	RMSNorm	Parallel	☑	RoPE	SwiGLU
Chinchilla	2022	RMSNorm	Serial	☑	Relative	ReLU
Mistral (7B)	2023	RMSNorm	Serial	☑	RoPE	SwiGLU
LLaMA2 (70B)	2023	RMSNorm	Serial	☑	RoPE	SwiGLU
LLaMA (65B)	2023	RMSNorm	Serial	☑	RoPE	SwiGLU
GPT4	2023			☐		
Baichuan 2	2023	RMSNorm	Serial	☑	AliBi	SwiGLU
Olmo 2	2024	RMSNorm	Serial	☐	RoPE	SwiGLU
Gemma 2 (27B)	2024	RMSNorm	Serial	☑	RoPE	GeGLU
Nemotron-4 (340B)	2024	LayerNorm	Serial	☑	RoPE	SqRelu
Qwen 2 (72b) - same for 2.5	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
Falcon 2 11B	2024	LayerNorm	Parallel	☑	RoPE	GeLU
Phi3 (small) - same for phi4	2024	RMSNorm	Serial	☑	RoPE	GeGLU
Llama 3 (70B)	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
Reka Flash	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
Command R+	2024	LayerNorm	Parallel	☑	RoPE	SwiGLU
OLMo	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
Qwen (14B)	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
DeepSeek (67B)	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
Yi (34B)	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
Mixtral of Experts	2024			☐		
Command A	2025	LayerNorm	Parallel	☑	Hybrid (RoPE+NoPE)	SwiGLU
Gemma 3	2025	RMSNorm	Serial	☐	RoPE	GeGLU
SmolLM2 (1.7B)	2025	RMSNorm	Serial	☑	RoPE	SwiGLU

Many variations in position embeddings

Sine embeddings: add sines and cosines that enable localization

$$Embed(x, i) = v_x + PE_{pos}$$

$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}})$$
$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}})$$

Notable models:

Original transformer

attention is position indepedent

Absolute embeddings: add a position vector to the embedding

$$Embed(x, i) = v_x + u_i$$

Notable models:

GPT1/2/3, OPT

Relative embeddings: add a vector to the *attention computation*

$$e_{ij} = \frac{x_i W^Q (x_j W^K + a_{ij}^K)^T}{\sqrt{d_z}}$$

Notable models:

T5, Gopher, Chinchilla

Rope embeddings (next slides..)

RoPE：目前最流行

Notable models:

GPTJ, PaLM, LLaMA

Most 2024+ models

RoPE: rotary position embeddings

High level thought process: a *relative* position embedding should be some $f(x, i)$ s.t.

$$\langle f(x,i), f(y,j) \rangle = g(x, y, i-j)$$

结果只取决于x和y之间的位置差异i-j

That is, the attention function *only* gets to depend on the relative position (i-j). How do existing embeddings not fulfill this goal?

- **Sine:** Has various cross-terms that are not relative

$$\langle Embed(x,i), Embed(y,i) \rangle = \langle v_x, v_y \rangle + \langle PE_i, v_y \rangle \ldots$$

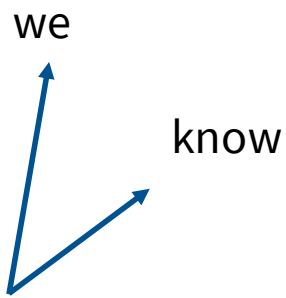
- **Absolute:** obviously not relative
- **Relative embeddings:** $e_{ij} = \frac{x_i W^Q (x_j W^K + a_{ij}^K)^T}{\sqrt{d_z}}$ is not an inner product

RoPE: rotary position embeddings

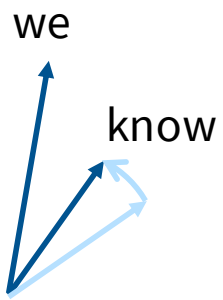
TODO：RoPE的详细底层原理理解

How can we solve this problem?

- We want our embeddings to be invariant to absolute position
- We know that inner products are invariant to arbitrary rotation.



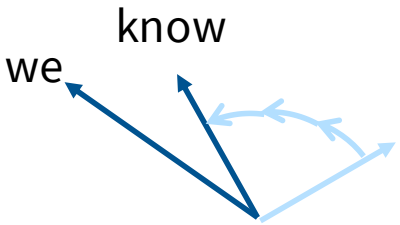
Position independent embedding



Embedding
“we know that”

Rotate we by ‘0 positions’

know by ‘1 positions’



Embedding
“of course we know”

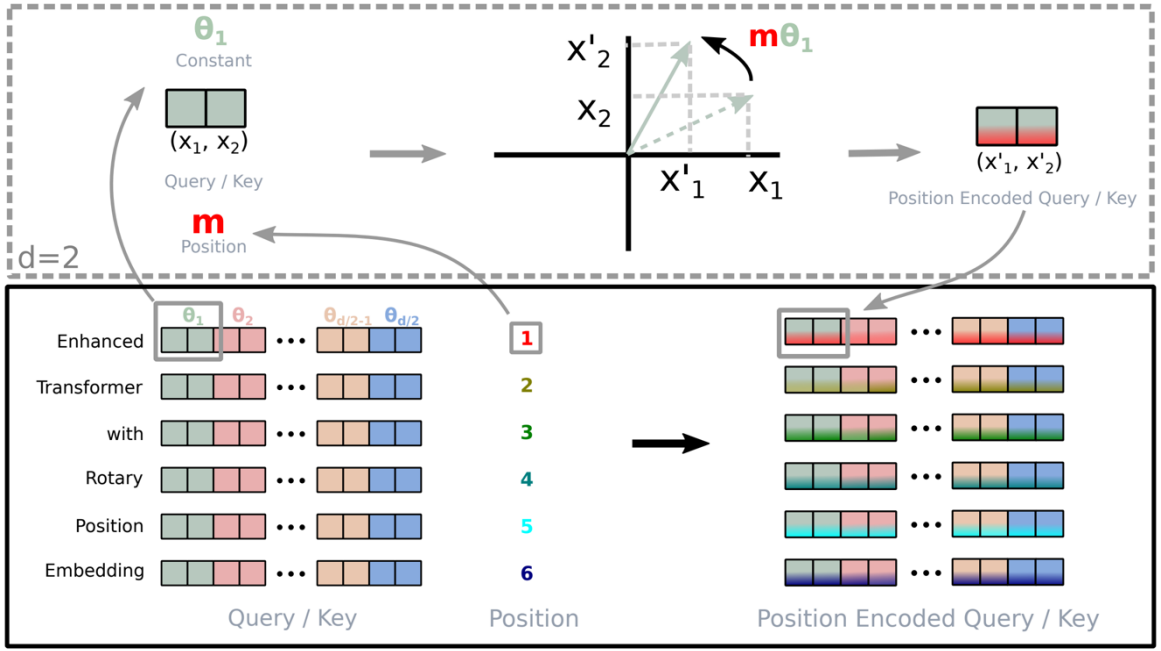
Rotate we by ‘2 positions’

Rotate know by ‘3 positions’

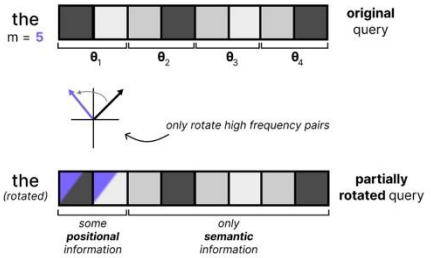
RoPE: rotary position embeddings

#TODO Gemma的p-RoPE和原本的RoPE的区别？改进点?

There are many rotations, which one do you pick?



[Su et al 2021]



Gemma 4 alternative: just first 2

D dimension：each pair都旋转

Just pair up the coordinates and rotate them in 2d (motivation: complex numbers)

The actual RoPE math

Multiply with sines and cosines

$$f_{\{q,k\}}(\boldsymbol{x}_m, m) = \boldsymbol{R}^d_{\Theta,m}\boldsymbol{W}_{\{q,k\}}\boldsymbol{x}_m \tag{14}$$

$$\boldsymbol{R}^d_{\Theta,m} = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix} \tag{15}$$

Difference with sine embeddings – not additive, no cross terms

Implementation and code for RoPE

Usual attention stuff

```
query_states = self.q_proj(hidden_states)
key_states = self.k_proj(hidden_states)
value_states = self.v_proj(hidden_states)

# Flash attention requires the input to have the shape
# batch_size x seq_length x head_dim x hidden_dim
# therefore we just need to keep the original shape
query_states = query_states.view(bsz, q_len, self.num_heads, self.head_dim).transpose(1, 2)
key_states = key_states.view(bsz, q_len, self.num_key_value_heads, self.head_dim).transpose(1, 2)
value_states = value_states.view(bsz, q_len, self.num_key_value_heads, self.head_dim).transpose(1, 2)
```

Get the RoPE matrix cos/sin

Multiply query/key inputs

```
cos, sin = self.rotary_emb(value_states, position_ids)
query_states, key_states = apply_rotary_pos_emb(query_states, key_states, cos, sin)
```

…

Same stuff as the usual multi-head self attention below

Note: embedding at *each attention operation* to enforce position invariance

#TODO 这个Note是啥意思?

Hyperparameters

Transformer hyperparameter questions you might have had in 224n..

- How much bigger should the feedforward size be compared to hidden size?
- How many heads, and should num_heads always divide hidden size?
- What should my vocab size be?

And other model setting questions

- Do people even regularize these huge LMs?
- How do people scale these models - very deep or very wide?

Surprising (?) consensus hyperparameter 1

Feedforward – model dimension ratio.

$$\mathrm{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

There are two dimensions that are relevant – the feedforward dim (d_{ff}) and model dim (d_{model}). What should their relationship be?

#TODO dff具体是啥?

$$\boldsymbol{d_{ff} = 4\, d_{model}}$$

This is *almost always* true. There's just a few exceptions.

这个4倍应该是经验之谈吧

Exception #1 – GLU variants

Remember that GLU variants scale down by 2/3rd. This means most GLU variants have $d_{ff} = \frac{8}{3} d_{model}$. This is mostly what happens. Some notable such examples.

Model	d_{ff}/d_{model}
PaLM	4
Mistral 7B	3.5
LLaMA-2 70B	3.5
LLaMA 70B	2.68
Qwen 14B	2.67
DeepSeek 67B	2.68
Yi 34B	2.85
T5 v1.1	2.5

Models are roughly in this range, though PaLM, LLaMA2 and Mistral are slightly larger

Exception #2 – T5

As we have (and will) see, most LMs are have boring, conservative hyperparameters. One exception is T5 [Raffel et al 2020] which has some *very bold* settings.

In particular, for the 11B model, they set

$$d_{ff} = 65{,}536$$
$$d_{model} = 1024$$

64倍，远远大于4倍

For an astounding 64-times multiplier.

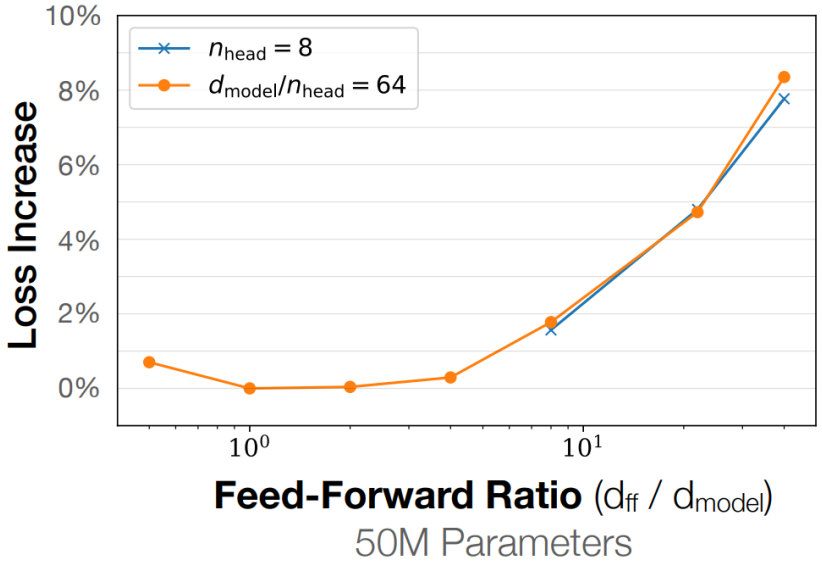
> for "11B" we use $d_{\text{ff}} = 65{,}536$ with 128-headed attention producing a model with about 11 billion parameters. We chose to scale up d_{ff} specifically because modern accelerators (such as the TPUs we train our models on) are most efficient for large dense matrix multiplications like those in the Transformer's feed-forward networks.

想法是64倍能够让GPU的运算更加有效率

Other, recent exceptions – Gemma 2 (8x), SmolLM/Gemma 3/Gemma 4 (4x, GLU)

Why this range of multipliers?

Empirically, there's a basin between 1-10 where this hyperparameter is near-optimal



[Kaplan+ 2020]

What can we learn from the model-dim hyperparam?

- The ‘default’ choices of $d_{ff} = 4d_{model}$ and $d_{ff} = 2.66d_{model}$ have worked well for nearly all modern LLMs.

- But T5 does show that even radical choices of $d_{ff} = 64d_{model}$ can work. This hyperparameter choice isn’t written in stone.

- That said, T5 has a follow-up model (T5 v1.1) that is ‘improved’ and uses a much more standard 2.5 multiplier on GeGLU, so the 64-times multiplier is likely suboptimal.

64倍的可能确实不是好方法

Surprising (?) consensus hyperparameter 2

#TODO 这页没看懂

Head-dim*num-heads to model-dim ratio. As a reminder, slide from 224n.

Multi-head self-attention is computationally efficient

- Even though we compute h many attention heads, it's not really more costly.
  - We compute $XQ \in \mathbb{R}^{n \times d}$, and then reshape to $\mathbb{R}^{n \times h \times d/h}$. (Likewise for XK, XV.)
  - Then we transpose to $\mathbb{R}^{h \times n \times d/h}$; now the head axis is like a batch axis.
  - Almost everything else is identical, and the **matrices are the same sizes.**

This doesn't *have to* be true: we can have head-dimensions > model-dim / num-heads.

But most models do follow this guideline

How many heads, whats the model dim?

Some examples of this hyperparameter

	Num heads	Head dim	Model dim	Ratio
GPT3	96	128	12288	1
T5	128	128	1024	16
T5 v1.1	64	64	4096	1
LaMDA	128	128	8192	2
PaLM	48	258	18432	1.48
LLaMA2	64	128	8192	1
Qwen 3.5 (27B)	24	256	5120	1.2

Most models have ratios around 1 – notable exceptions by some google models.

Aspect ratios

#TODO 模型的aspect ratio指的是什么?

Should my model be deep or wide? *How* deep and how wide?

Most models are surprisingly consistent on this one too!

Model	d_{model}/n_{layer}
BLOOM	205
T5 v1.1	171
PaLM (540B)	156
GPT3/OPT/Mistral/Qwen /OLMo 3	128
LLaMA / LLaMA2	102
Gemma 3	87
Gemma 4	61
T5 (11B)	33

Sweet spot?

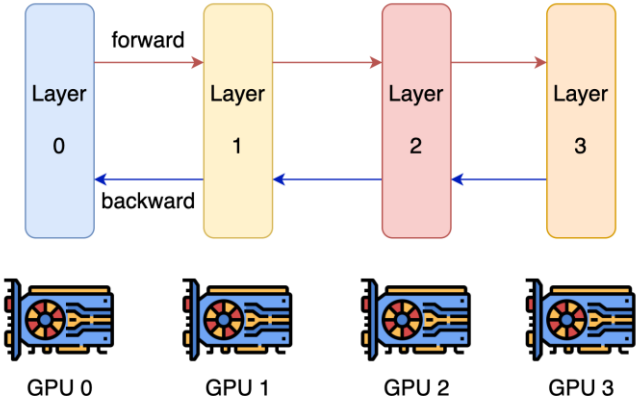
这个比例一般不会差别太大

Considerations about aspect ratio

Extremely deep models are harder to parallelize and have higher latency

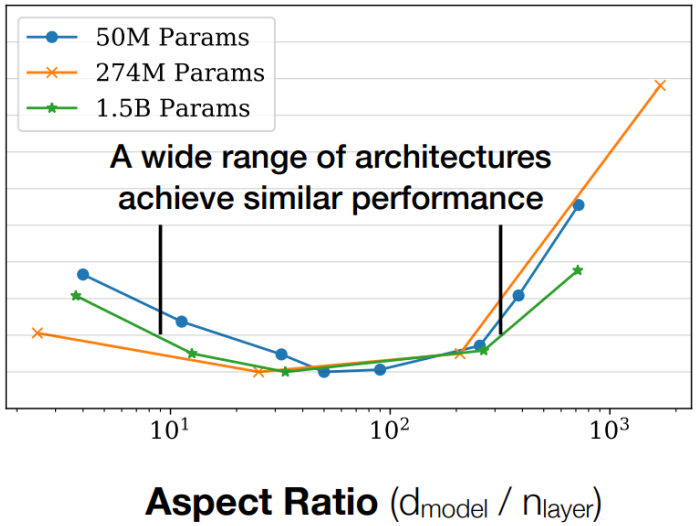
> **The Limits of Depth vs Width** We note an obvious limitation with our advice. Scaling depth has an obvious limiter, i.e., they are non-parallelizable across different machines or devices and every computation has to always wait for the previous layer. This is unlike width, which can be easily parallelizable over thousands or hundreds of thousands of devices. Within the limitation of scaling

[Tay et al 2021]

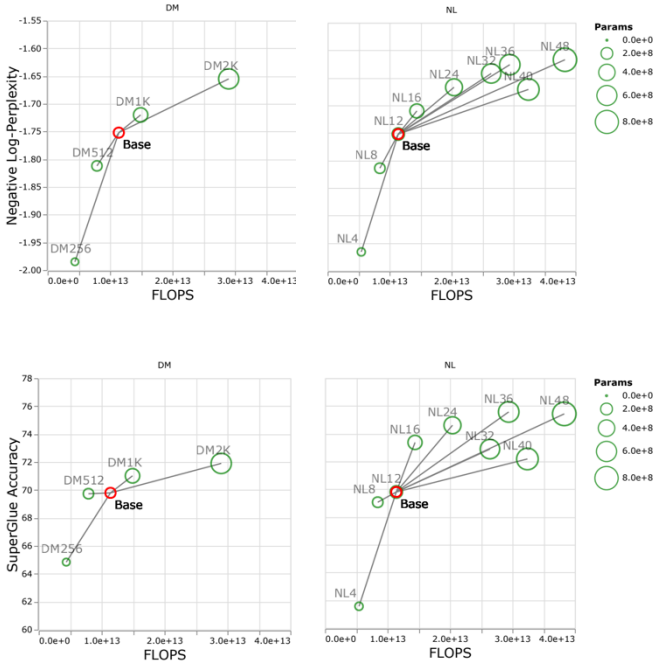


在wide维度scale，相比在deep维度scale简单很多，也更好组织

Evidence on aspect ratio scaling



[Kaplan et al 2020]



[Tay et al 2021]

What are typical vocabulary sizes?

#TODO 词表大小会影响什么？这两张图中区别是啥？

Monolingual models – 30-50k vocab

Model	Token count
Original transformer	37000
GPT	40257
GPT2/3	50257
T5/T5v1.1	32128
LLaMA	32000

Multilingual / production systems 100-250k

Model	Token count
mT5	250000
PaLM	256000
GPT4	100276
Gemma 4	262144
DeepSeek	100000
Qwen 15B	152064
Yi	64000

Monolingual vocabs don’t need to be huge, but multilingual ones do

Dropout and other regularization

Do we need regularization during pretraining?

dropout就是一种正则化手段

Arguments against:

- There is *a lot* of data (trillions of tokens), more than parameters.
- SGD only does a single pass on a corpus (hard to memorize)

不需要正则化的观点：一般来说，我们训练时候有着充足的数据，所以我们一般只做one-epoch，应该是这个意思
因此，对于我们的data，SGD只会经过一次。所以，根本不存在overfitting的问题，也就不用regularization了

This is all quite reasonable.. but what do people do in practice?

我理解是当前的很多模型，相比参数量，数据还是太多了。不容易过拟合，也就不用正则化了。

Dropout and weight decay in practice

Model	Dropout*	Weight decay
Original transformer	0.1	0
GPT2	0.1	0.1
T5	0.1	0
GPT3	0.1	0.1
T5 v1.1	0	0
PaLM	0	(variable)
OPT	0.1	0.1
LLaMA	0	0.1
Qwen 14B	0.1	0.1

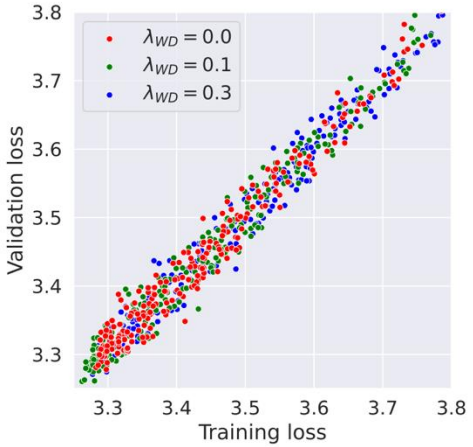
Many older models used dropout during pretraining

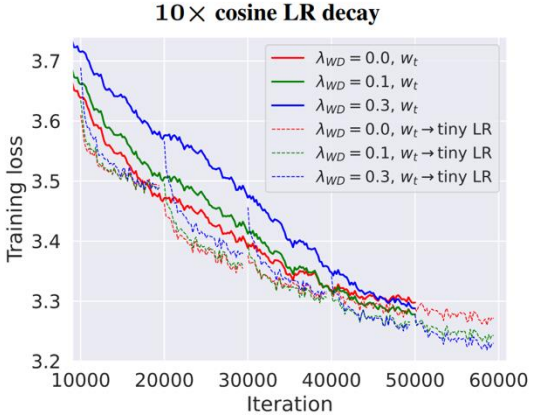
Newer models (except Qwen) rely only on weight decay

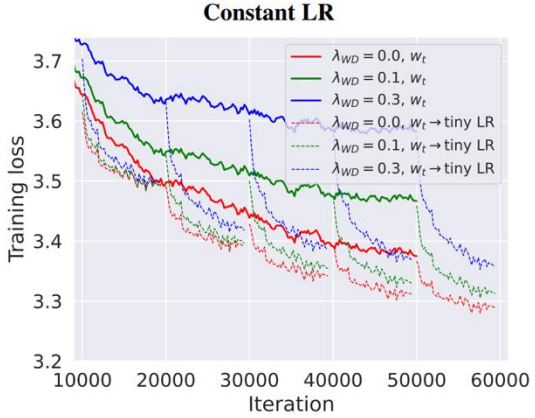
* Most of the times papers just don't discuss dropout. On open models, this closely matches not doing dropout. This may not be true of closed models.

Why weight decay LLMs?

[Andriushchenko et al 2023] has interesting observations about LLM weight decay







It's not to control overfitting

Weight decay interacts with learning rates (cosine schedule)

#TODO：cosine LR decay的原理，以及weight decay和学习率是交互的，这是啥意思?

Summary: hyperparameters

Feedforward

- Factor-of-4 rule of thumb (8/3 for GLUs) is standard (with some evidence)

Head dim

- Head dim*Num head = D model is standard – but low to no validation

Aspect ratio

- Wide range of ‘good’ values (100-200). Systems concerns dictate the value

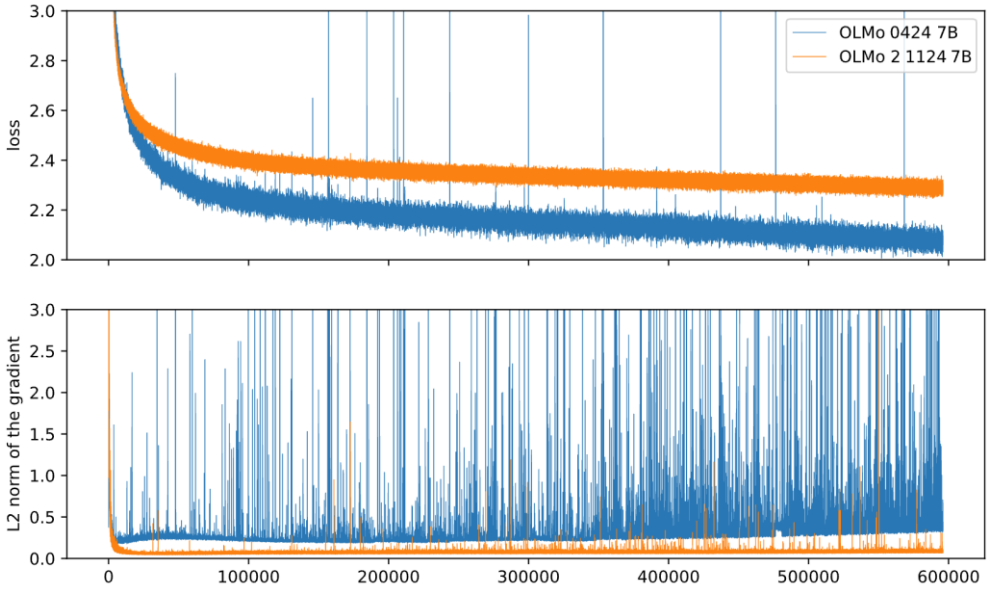
Regularization

- You still ‘regularize’ LMs but its effects are primarily on optimization dynamics

Name	Year	MLP factor	Aspect ratio (d/layer)	weight decay	drop_rate
Original transformer	2017	4	85	0	0.1
GPT	2018	4	64	0.1	0.1
T5 (11B)	2019	64	43	0	0.1
GPT2	2019	4	33	0.1	0.1
T5 (XXL 11B) v1.1	2020	2.5	171	0	0
mT5	2020	2.5	171	0	0
GPT3 (175B)	2020	4	128	0.1	0.1
GPTJ	2021		146	0.1	0
LaMDA	2021	8	128		
Anthropic LM (not claude)	2021	4	128		
Gopher (280B)	2021	4	205		
GPT-NeoX	2022	4	140	0.01	0
BLOOM (175B)	2022	4	205	0.1	0
OPT (175B)	2022	4	128	0.1	0.1
PaLM (540B)	2022	4	156		0
Chinchilla	2022	4	102		
Baichuan 2	2023	2.68	128	0.1	0
Mistral (7B)	2023	3.5	128	0.1	0
LLaMA2 (70B)	2023	3.5	102	0.1	0
LLaMA (65B)	2023	2.6875	102	0.1	0
GPT4	2023		0		
Olmo 2	2024	2.6875	128		
Gemma 2 (27B)	2024	8	100		
Nemotron-4 (340B)	2024	4	192		0
Qwen 2 (72b) - same for 2.5	2024	3.609	102		
Falcon 2 11B	2024	4	68	0.1	
Phi3 (small) - same for phi4	2024	3.5	128		
Llama 3 (70B)	2024	3.5	102		0
Reka Flash	2024		0		
Command R+	2024	2.75	192		
OLMo	2024	2.6875	128	0.1	0
Qwen (14B)	2024	2.675	128	0.1	0.1
DeepSeek (67B)	2024	2.6875	86	0.1	0
Yi (34B)	2024	2.857142	119	0.1	0
Mixtral of Experts	2024		0		
Command A	2025		0		
Gemma 3	2025	4	87		
SmolLM2 (1.7B)	2025	4	85		

Stability tricks

Recently, lots of attention on *stable training*

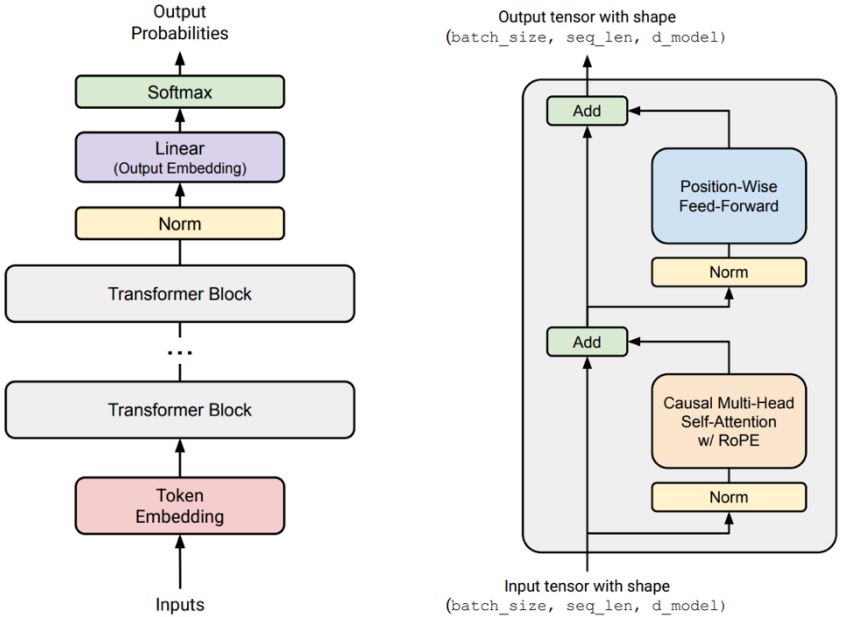


训练稳定性很重要

Don’t train models that look like the blue curve!

Where do the issues arise? Beware of softmaxes!

Softmaxes – can be ill-behaved due to exponentials / divison by zero



Output softmax stability – the ‘z-loss’

Recall the softmax calculation

$$\begin{aligned} \log(P(x)) &= \log\left(\frac{e^{U_r(x)}}{Z(x)}\right) \\ &= U_r(x) - \log(Z(x)) \\ Z(x) &= \Sigma_{r'=1}^{|V|} e^{U_{r'}(x)} \end{aligned}$$

zloss: 增加对于log(Z)偏离0的惩罚

$$\begin{aligned} L &= \sum_i \left[\log(P(x_i)) - \alpha(\log(Z(x_i)) - 0)^2\right] \\ &= \sum_i \left[\log(P(x_i)) - \alpha \log^2(Z(x_i))\right] \end{aligned}$$

[From Devlin 2014]

第一项U一般不会上下溢出，比较稳定；第二项Z则可能很不稳定，因为是指数运算

This is useful for stability! PaLM used this ‘z loss’ trick.

> We additionally use an auxiliary loss of $z_loss = 10^{-4} \cdot \log^2 Z$ to encourage the softmax normalizer $\log(Z)$ to be close to 0, which we found increases the stability of training.

Other examples: Baichuan 2 (2023), DCLM (2024), OLMo 2 (2025), OLMo 3 (2025)

Attention softmax stability – the ‘QK norm’

在QK^T的运算之前先对Q和K都做一次Layernorm



The query and keys ar softmax operation.

Other examples: DCLM, OLMo2, Gemma 2, Qwen3, OLMo 3, Gemma 4

Originally from vision and multimodal models [Dehgani 2023, Idefcs, Chameleon]

Logit soft-capping.

Soft-capping the logits to some maximum value via Tanh

大概是直接对logits做上下界clip

Logit soft-capping. We cap logits (Bello et al., 2016) in each attention layer and the final layer such that the value of the logits stays between −soft_cap and +soft_cap. More specifically, we cap the logits with the following function:

$$\text{logits} \leftarrow \text{soft_cap} * \tanh(\text{logits}/\text{soft_cap}).$$

We set the soft_cap parameter to 50.0 for the self-attention layers and to 30.0 for the final layer.

Prevents logits from blowing up, but also might have perf issues?

稳定了但是可能会降低表现

Table 4: Models perplexity with confidence interval ±0.1 at 95% level.

bf16 baseline	*soft_cap*	*QKV_norm*	*QK_norm_cap*	*QK_norm*	*QK_FC_norm*
11.19	11.24	10.85	11.00	10.84	10.87

Attention heads

Most models don’t touch the attention heads much at all with a few minor exceptions..

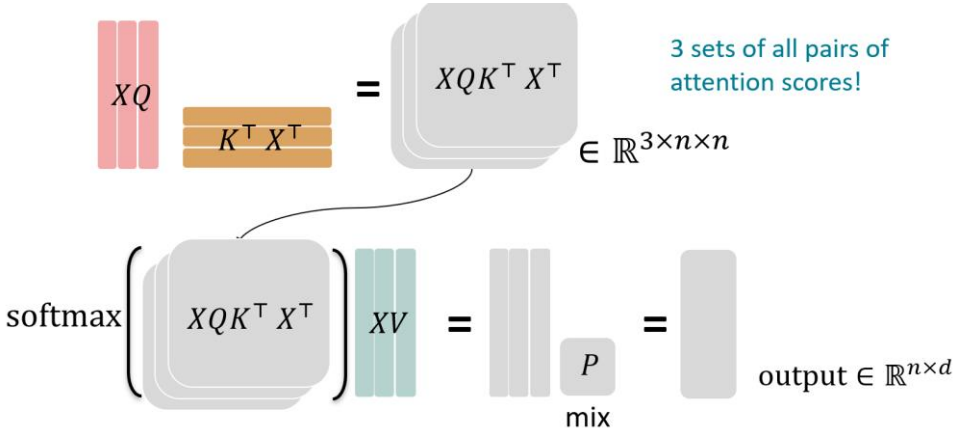
GQA / MQA : Saving inference costs by reducing the number of heads

Sparse or sliding window attention (GPT4/Mistral): restricting the attention pattern to reduce compute cost

Exotic SSM stuff (Jamba, Falcon 3, Qwen 3.5, etc): next lecture!

GQA/MQA – Reducing attention head cost

Let's think about the compute involved for attention



d = hidden dim
b = batch
n = length (<d)
h = heads
k = head dim (d/h)

X softmax projection

Total arithmetic operations (bnd^2), **total memory accesses** $(bnd + bhn^2 + d^2)$

#TODO 这里的arithmetic intensity是如何计算出来的?

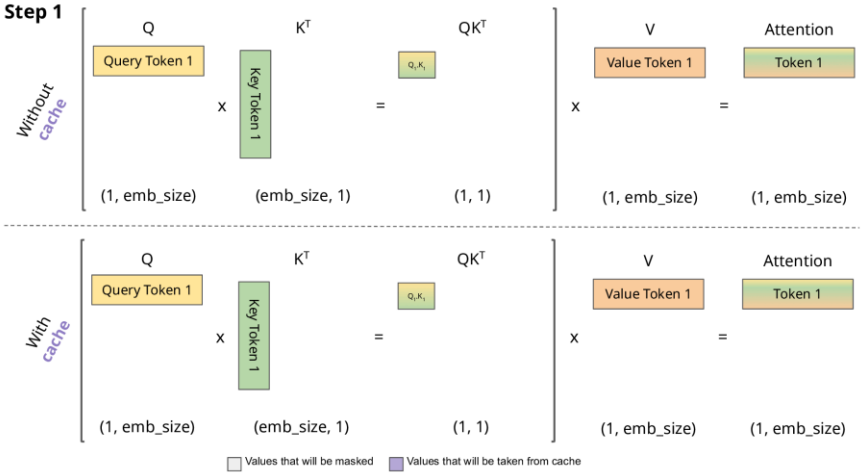
Arithmetic intensity (compute/memory) is high $O\left(\left(\frac{1}{k}+\frac{1}{bn}\right)^{-1}\right)$ - we can keep our GPUs running

GQA/MQA – Reducing attention head cost

What about the *incremental* case when we generate text?

Key difference: can't parallelize the generation process – needs to be step by step

In this case – we need to incrementally re-compute/update attention via the 'KV cache'



[Animation from https://medium.com/@joaolages/kv-caching-explained-276520203249]

GQA/MQA – Reducing attention head cost

#TODO 这页的含义和逻辑具体是什么？我不理解。

What's the incremental arithmetic intensity?

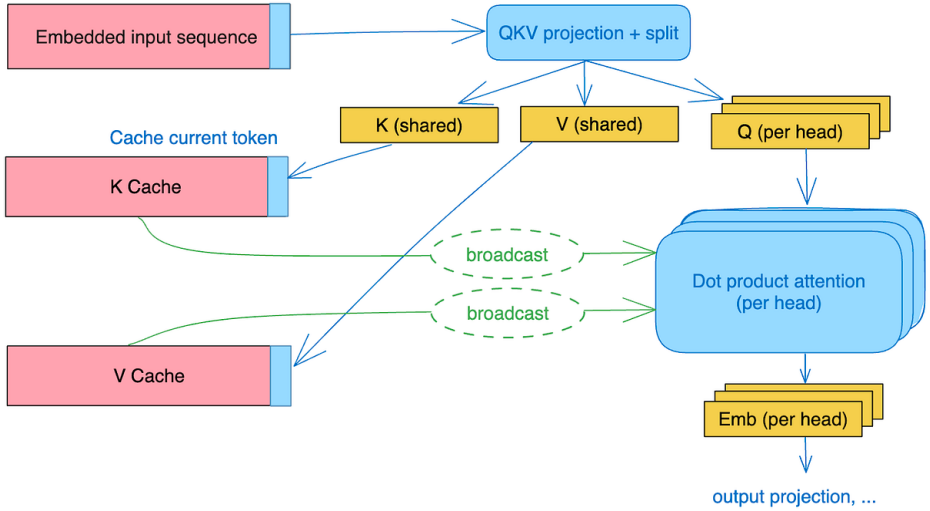
Total arithmetic operations (bnd^2), **total memory accesses** $(\overset{\text{K, V}}{bn^2d} + \overset{\text{projection}}{nd^2})$

Arithmetic intensity is not good $O\left(\left(\frac{n}{d} + \frac{1}{b}\right)^{-1}\right)$ - need large batches + short seq length (n) or big model dimensions (d)

Is there some way around this? The n/d term is difficult to reduce.

MQA – just have fewer key dimensions.

Key idea – have multiple queries, but just one dimension for keys and values



所有head使用同样的K和V，只有Q不同

k=d/h，代表头数head_num

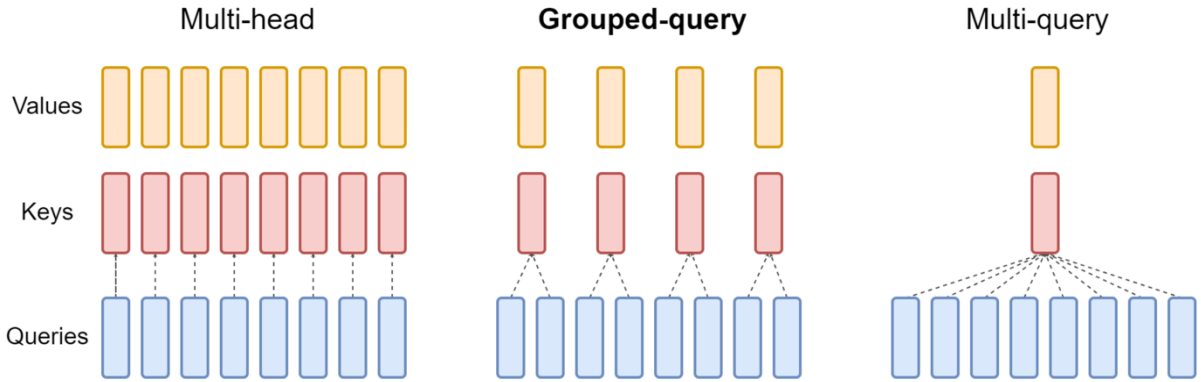
We have much fewer items to move in and out of memory (KV Cache)

Total memory access $(bnd + bn^2k + nd^2)$, **Arithmetic intensity** $O\left(\left(\frac{1}{d} + \frac{n}{dh} + \frac{1}{b}\right)^{-1}\right)$

[figure from https://blog.fireworks.ai/multi-query-attention-is-all-you-need-db072e758055]

Additional extensions – GQA

Don’t go all the way to one dimension of KV – have fewer dims



Simple knob to control expressiveness (key-query ratio) and inference efficiency

More recently – MLA (multihead latent attention) from deepseek v2

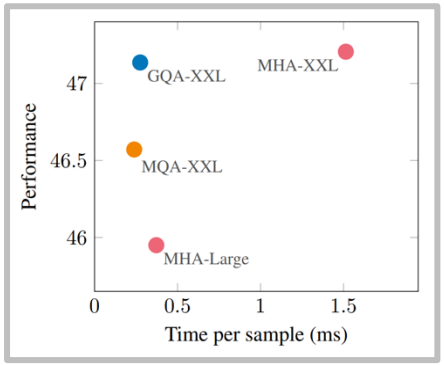
Does MQA hurt? Sometimes..

Small PPL hit w/ MQA [Shazeer 2019]

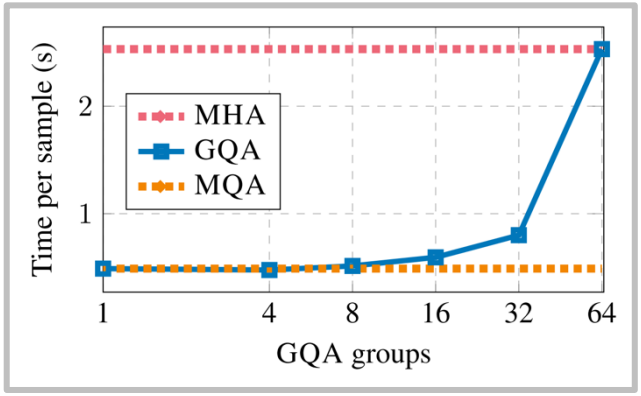
Table 3: Billion-Word LM Benchmark Results.

Attention	h	d_k, d_v	d_{ff}	dev-PPL
multi-head	8	128	8192	**29.9**
multi-query	8	128	9088	30.2
multi-head	1	128	9984	31.2
multi-head	2	64	9984	31.1
multi-head	4	32	9984	31.0
multi-head	8	16	9984	30.9

Low/no hit w/ GQA [Ainslie 2023]



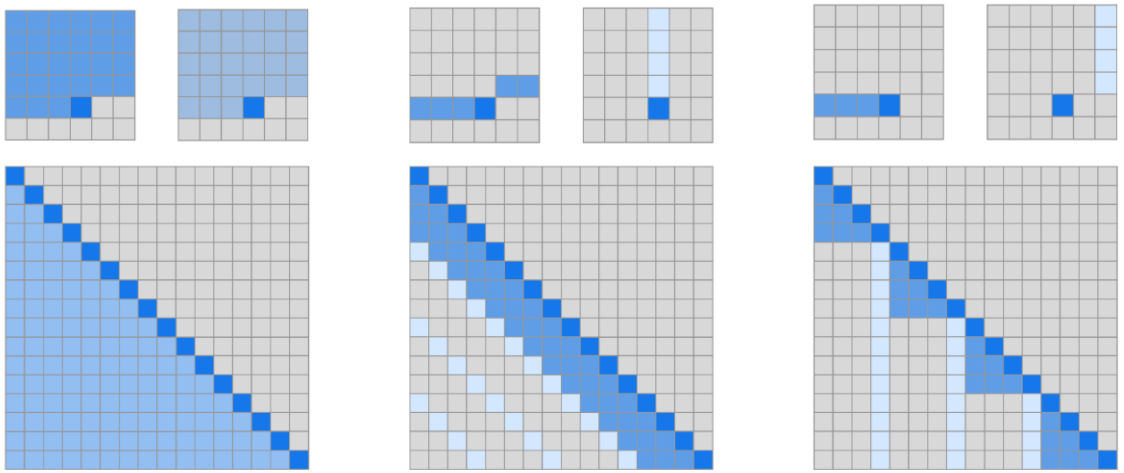
GQA看着最好



Sparse / sliding window attention

Attending to the entire context can be expensive (quadratic).

Build sparse / structured attention that trades off expressiveness vs runtime (GPT3, GPT-OSS, Gemma4)



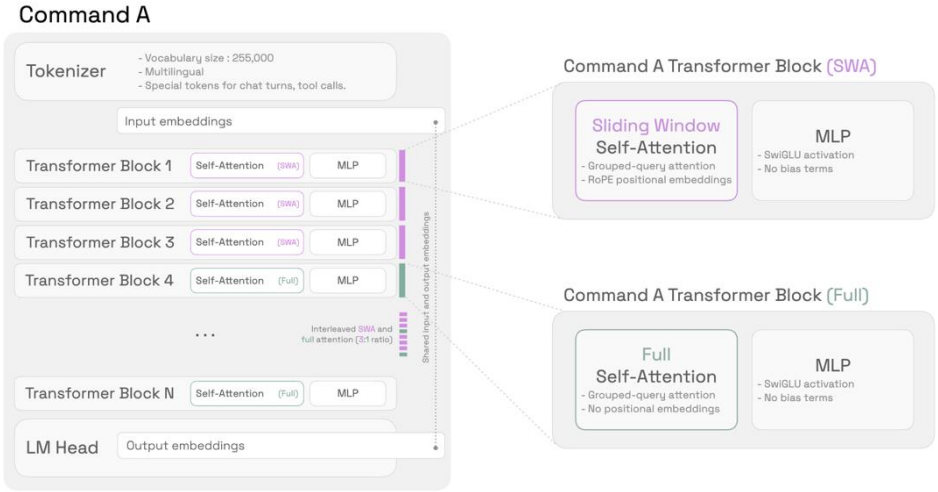
(a) Transformer (b) Sparse Transformer (strided) (c) Sparse Transformer (fixed)

[Child et al 2019]

Current standard trick – interleave ‘full’ and ‘LR’ attention

From Cohere Command A – Every 4th layer is a full attention

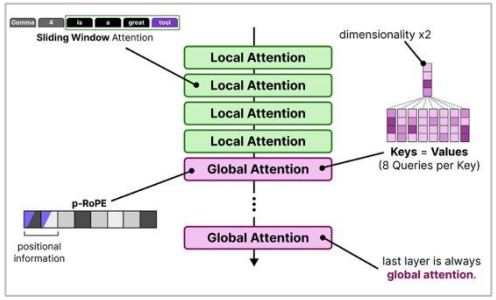
交替full-attn和sw-attn
每四个，有一个是full-attn



Long-range info via NoPE, short-range info via RoPE + SWA.

Other models – LLaMA 4, Gemma 3, Gemma 4, OLMo 3 does SWA+Full RoPE.

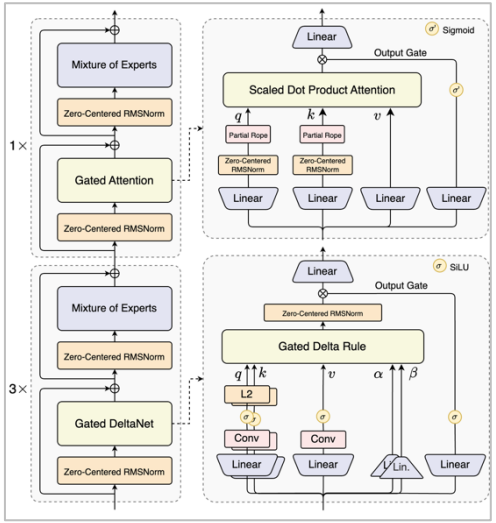
Other recent examples of interleaved attention



Gemma 4

Gradient clipping	1.0
Z-loss weight	10^{-5}
Weight decay on embeddings	No
Sliding window attention	3/4 of layers; 4,096 tokens
RoPE scaling	YaRN on full attn. layers
RoPE θ	$5 \cdot 10^5$
Layer norm applied to	Outputs

Olmo 3

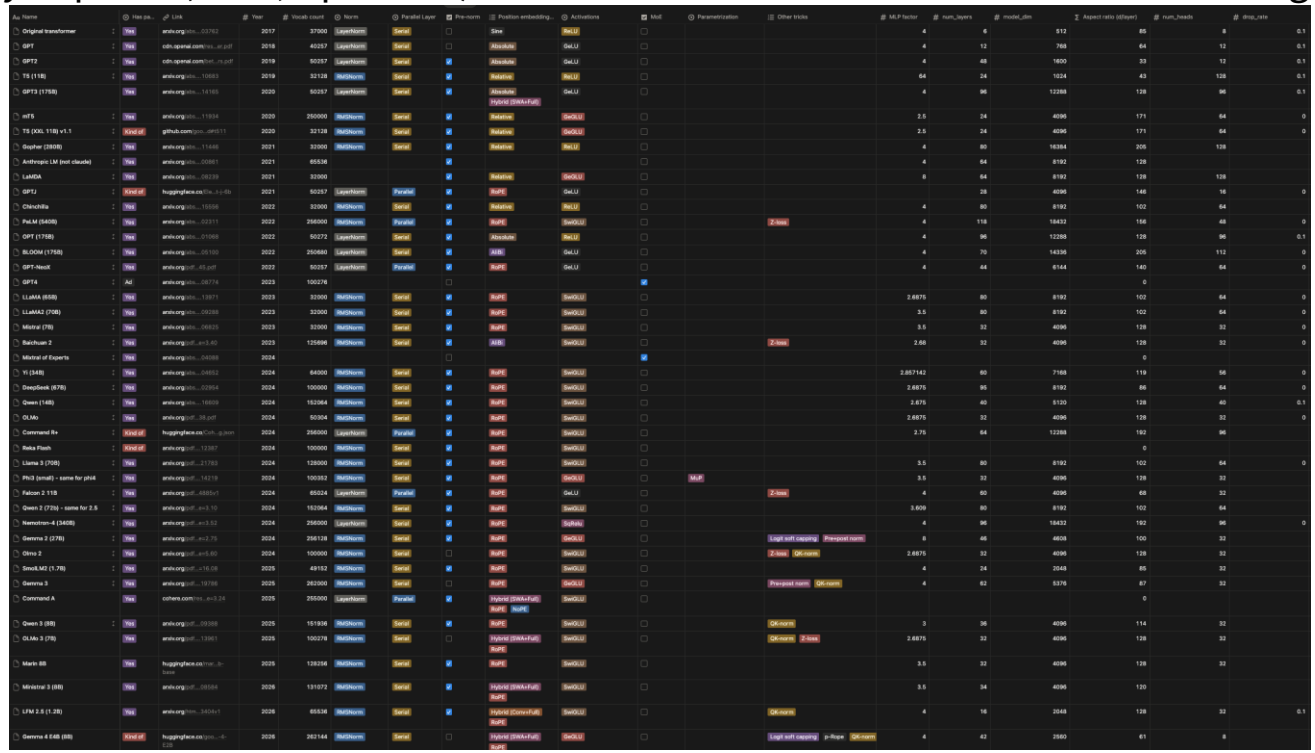


Qwen 3.5 / Qwen 3 Next

https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-gemma-4

Recap, conclusion, etc.

Many aspects (arch, hparams) of transformers are in common across the big LMs



Major differences? Position embeddings, activations, tokenization